CM-IDS-001 Interior Component Design Standard

Revision C | Effective 2026-03-01 | Owner: Interior Engineering Standards Group

1. Scope

This standard defines the minimum requirements every interior component specification must meet before release. It applies to instrument panel, door trim, console, seat foam, headliner and pillar trim components. Each released specification is checked against Sections 2 to 8. The overall validation result is the most severe result of any single check (Section 9).

2. Approved Materials

Only the materials below may be used for each component category. Any other material requires a deviation approved by the Engineering Review Board and is otherwise a FAIL.

Component category	Approved materials
Instrument Panel	PP-TD20, PC/ABS
Door Trim	PP-TD20, PP-GF20
Console	ABS, PC/ABS
Seat Foam	PU Foam
Headliner	PET Fiber Composite
Pillar Trim	PP-TD20, TPO

3. Mass Targets

Actual mass is compared with the target mass recorded on the specification.

Actual / target	Result
Up to 105%	PASS
Above 105% up to 110%	WARNING
Above 110%	FAIL

4. Cost Targets

Actual piece cost (USD) is compared with the target piece cost.

Actual / target	Result
Up to 103%	PASS
Above 103% up to 108%	WARNING
Above 108%	FAIL

5. Flammability

All interior materials must meet FMVSS 302 (burn rate not more than 102 mm/min). The internal design target is 80 mm/min or less to keep margin for supplier variation.

Burn rate (mm/min)	Result
80 or less	PASS
Above 80 up to 102	WARNING
Above 102	FAIL (does not meet FMVSS 302)

6. Odor

Odor is rated with VDA 270 on a 1 to 6 scale (lower is better).

VDA 270 rating	Result
3.0 or less	PASS
3.5	WARNING
4.0 or higher	FAIL

7. Seat Foam Density

Seat foam density must be between 45 and 60 kg/m3 (inclusive). Outside this range is a FAIL because of durability and comfort risk.

8. Documentation

Every specification must include: part number, revision letter, component category, material, target and actual mass, target and actual cost, flammability result, odor rating, supplier and engineer approval. A specification marked DRAFT, or without engineer approval, is a WARNING and cannot be released until corrected.

9. Validation Outcome and Escalation

The overall result is the most severe individual result (FAIL over WARNING over PASS).

Overall result	Required action
PASS	Specification may be released.
WARNING	Responsible engineer adds a written justification within 3 business days.
FAIL	Escalate to the Engineering Review Board within 5 business days.

10. Contacts

Standard owner: Interior Engineering Standards Group. Engineering Review Board meets every Tuesday. Deviation requests are submitted through the Interior Engineering SharePoint site.

Revision History

Rev	Date	Change
A	2024-09-15	Initial release
B	2025-05-02	Added VDA 270 odor requirement
C	2026-03-01	Tightened cost FAIL threshold from 110% to 108%